

Reduction #9: The Gabor-kernel Tempotron

Band-pass Radiality and the Two-Quadrature Capacity Bound

Tempotron learning-surface program

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Abstract

We treat the Gabor (band-pass) post-synaptic-potential (PSP) kernel $K(s) = e^{-s^2/2\sigma^2} \cos(\omega_0 s)$ as a stand-alone reduction: the program’s *radiality axis*. Its quality factor $Q = \omega_0 \sigma$ interpolates continuously from the low-pass Gaussian ($Q \rightarrow 0$) to a single carrier frequency ($Q \rightarrow \infty$); the effective dimension is pinned exactly by the spectral moments,

$$K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0} = T\sqrt{\omega_0^2 + \frac{1}{2\sigma^2}} = \omega_0 T\sqrt{1 + \frac{1}{2Q^2}}.$$

Our main contribution is theoretical. In the narrowband limit the membrane potential collapses to a single quadrature pair, $V(t) = A \cos \omega_0 t + B \sin \omega_0 t$ with $A = \mathbf{w} \cdot \mathbf{a}$, $B = \mathbf{w} \cdot \mathbf{b}$, so that $\max_t V = \sqrt{A^2 + B^2}$ and the tempotron decision becomes the *two-quadrature radial classifier* $\sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2} > \theta$. Solving the replica-symmetric (RS) Gardner distance-to-feasible-set problem for this geometry gives a closed form,

$$\frac{1}{\alpha_c(\theta)} = \frac{1}{2}(\theta^2 - \sqrt{2\pi}\theta + 2),$$

which at the balanced (median) threshold $\theta = \sqrt{2 \ln 2}$ yields $\boxed{\alpha_c^{\text{RS}} = 4.598}$ (4.660 threshold-optimized). This is, to our knowledge, a *new* value: the two-quadrature norm-threshold capacity is unpublished, and it is decisively distinct from the literature one-quadrature squared-perceptron value $\alpha_c = 4$ (Gratsea–Kasper–Lewenstein; Cruciani). We verified it four independent ways (a Gardner min-cost calculation, a SymPy closed form, an 8 M-sample Monte-Carlo 4.601, and an independent codex re-derivation), all anchored to the exact linear value 2. Because the fire region is the *exterior of a disk*—non-convex—RSB is permitted, so 4.598 is an RS/existence value and the findable capacity is lower. On the experimental side we report honestly: the narrowband Gabor crosses at $\hat{\alpha}_c \approx 3.55$ (the radial regime, findable well below the RS 4.6, consistent with the single-sinusoid reduction #6 at 3.4–3.8), but the *matched- K_{eff}* band-pass>low-pass isolation could *not* be re-measured here because the *v14* Gabor K_{eff} -targeting had a grid-sizing inconsistency (all cells railed to $K_{\text{eff}} \approx 499$); that comparison rests on the prior cross-kernel result (reduction #7: Gabor +0.3 to +0.6 over Gaussian at matched K_{eff} , reaching 3.89). Status: **PARTIAL**—the radial theory is solid and quadruply verified; the matched- K_{eff} control is an honest open gap.

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Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

1 Introduction

The tempotron [8] fires when its peak membrane potential exceeds threshold,

$$V_{\max}(\mathbf{w}) = \max_{t \in [0, T]} \sum_{i=1}^N w_i \sum_f K(t - t_i^f), \quad (1)$$

where $K(s)$ is the PSP kernel and $\{t_i^f\}$ the presynaptic spikes. The companion reductions in this series isolate one structural property of K at a time: smoothness (#8 Gaussian, #11 rectangular), causality (#10 alpha), band-limitedness (#6 sinusoidal). The present paper isolates *radiality*: what happens to capacity when the kernel carries a frequency, so the readout sees not a single linear projection but the *magnitude* of a two-dimensional projection.

Why the Gabor kernel. The Gabor elementary signal [5]

$$K(s) = e^{-s^2/2\sigma^2} \cos(\omega_0 s), \quad Q := \omega_0 \sigma, \quad (2)$$

is the minimal kernel whose quality factor Q dials *continuously* between two qualitatively different readouts. At $Q \rightarrow 0$ the cosine flattens and $K \rightarrow e^{-s^2/2\sigma^2}$, the pure low-pass Gaussian of reduction #8: a smooth broadband process read out by a best-of-many-looks linear perceptron. At $Q \rightarrow \infty$ the envelope is flat over any finite spike window and $K \rightarrow \cos \omega_0 s$, a single carrier: the readout becomes the amplitude of one in-phase/quadrature pair, a categorically non-linear decision. The Gabor kernel is therefore the one that *breaks* a naive “ α_c is a function of K_{eff} alone” universality and forces a two-axis picture (effective dimension $\times$ radiality, the subject of the cross-kernel reduction #7).

Contribution. We (i) write the exact narrowband reduction of the tempotron decision to a two-quadrature radial classifier (§2); (ii) *solve* the RS capacity of that classifier in closed form and obtain a new value $\alpha_c^{\text{RS}} = 4.598$, distinguishing it sharply from the literature one-quadrature $\alpha_c = 4$ and verifying it four ways (§3); (iii) report the findable lab capacity ($\hat{\alpha}_c \approx 3.55$, well below the RS value, as the non-convexity predicts) and state plainly the one experimental control we could not re-run (§4); and (iv) lay out the honest status—what is RS-level, where RSB and finite-window anisotropy enter, and what remains open (§5).

2 Model and Setup

Model. N afferents emit spike trains over $[0, T]$; the membrane potential is (1) with the Gabor kernel (2). Classification is $V_{\max} > \theta \Rightarrow \text{fire}$, on balanced random ± 1 labels; the capacity $\alpha_c = P/N$ is the half-solving load. The linear anchor is Cover’s $\alpha_c = 2$ [3, 6, 10].

Assumptions ledger.

- (A1) **Gaussian field.** For $N \gg 1$ afferents the CLT makes $V(t)$ a stationary zero-mean Gaussian process with covariance $\mathcal{C}(\tau) = (K \star K)(\tau)$ and spectrum $S(\omega) = |\hat{K}(\omega)|^2$ (Wiener–Khinchin). This is the same $N \gg 1$ premise as reductions #3, #4, #6; it can break under sparse input / finite- N starvation.
- (A2) **Narrowband reduction.** As $Q \rightarrow \infty$ the decision statistic $\max_t V$ reduces *exactly* to the radial form $\sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2} > \theta$ (§3.1); at finite Q extra modes and the envelope make the two quadratures correlated and unequal-power (§3.5).
- (A3) **Isotropic quadratures.** In that limit the two quadrature fields $h_1 = \mathbf{w} \cdot \mathbf{a} / \sqrt{N}$, $h_2 = \mathbf{w} \cdot \mathbf{b} / \sqrt{N}$ are i.i.d. $\mathcal{N}(0, 1)$ for fixed $\mathbf{w}$ on the sphere (uniform phase, $\omega_0 T \gg 1$). Finite T / non-uniform phase introduces an anisotropy $\rho > 0$ (§3.5, the load-bearing finite-size correction).
- (A4) **RS Gardner min-cost.** We use the RS, $q \rightarrow 1$ (zero-error) distance-to-feasible-set capacity (§3.2); it is *exact at the linear anchor* (reproduces 2). The exterior-of-disk region is non-convex, so RSB is permitted and would *lower* α_c .
- (A5) **Balanced threshold.** θ is the median of $\max_t V$ (per-pattern fire probability $\frac{1}{2}$), the operating point the lab uses and the convention under which the $\ln \ln K$ cluster argument of reduction #3 holds. A *margin- κ* convention (the literature $\alpha_c = 4$) is a *different* problem (§3.3).

Every *absolute* α_c below is RS-level; the robust outputs are the *value* of the radial limit (and its provenance), the *sign* of the band-pass advantage, and the *shape* of $\alpha_c(Q)$.

Exact effective dimension. For the Gabor spectrum (two Gaussian bumps at $\pm \omega_0$ of width $1/\sigma$), the spectral moments give

$$\frac{\lambda_2}{\lambda_0} = \omega_0^2 + \frac{1}{2\sigma^2} \implies \boxed{K_{\text{eff}} = T \sqrt{\omega_0^2 + \frac{1}{2\sigma^2}} = \omega_0 T \sqrt{1 + \frac{1}{2Q^2}}.} \quad (3)$$

The narrowband asymptote $\omega_0 T (1 + \frac{1}{4Q^2})$ already *under-counts* by $\sim 3\%$ at $Q = 2$ relative to the exact $\sqrt{1 + 1/2Q^2}$, so a cross-kernel comparison must match cells on the *exact* axis (3) or it inherits a Q -dependent bias that would mimic a radiality effect. The Gaussian ($Q \rightarrow 0$) limit recovers $K_{\text{eff}} = T/(\sigma\sqrt{2})$ of reduction #8.

3 Theory: The Two-Quadrature Radial Classifier

This section is the main contribution.

3.1 Exact narrowband reduction

As $Q \rightarrow \infty$ the envelope $e^{-s^2/2\sigma^2}$ is flat across any finite spike window, and $K(s) \rightarrow \cos \omega_0 s = \cos \omega_0 t \cos \omega_0 t_s + \sin \omega_0 t \sin \omega_0 t_s$. Substituting into (1) and collecting terms,

$$V(t) \longrightarrow A \cos \omega_0 t + B \sin \omega_0 t, \quad A = \mathbf{w} \cdot \mathbf{a}, \quad B = \mathbf{w} \cdot \mathbf{b}, \quad (4)$$

with the cos/sin quadrature features

$$a_i = \sum_f \cos \omega_0 t_i^f, \quad b_i = \sum_f \sin \omega_0 t_i^f. \quad (5)$$

A single sinusoid attains its maximum over t at its amplitude, so

$$\max_t V(t) = \sqrt{A^2 + B^2} = \sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2}. \quad (6)$$

Hence the firing rule is

$$\text{fire} \iff (\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2 > \theta^2 \iff \mathbf{w}^\top M \mathbf{w} > \theta^2, \quad M = \mathbf{a}\mathbf{a}^\top + \mathbf{b}\mathbf{b}^\top \text{ (rank 2)}. \quad (7)$$

The decision surface in weight space is a quadric—the *exterior* of a rugby-ball ellipsoid—categorically non-linear. Because M is rank-2 rather than a free $N \times N$ form, the classifier keeps only N free parameters: its capacity is $O(1)$, not the extensive-in- N capacity of a full quadratic gate (Baldi–Vershynin [1]). Geometrically the decision lives in the $(A, B) = (\mathbf{w} \cdot \mathbf{a}, \mathbf{w} \cdot \mathbf{b})$ plane: fire is the exterior of a disk of radius θ , silent its interior (Fig. 1a).

3.2 RS Gardner distance-to-feasible-set capacity

We compute the zero-error RS capacity in the $q \rightarrow 1$ limit through the Gardner “min-cost” representation: the inverse capacity equals the expected squared distance from a frozen Gaussian field to the label-allowed region,

$$\frac{1}{\alpha_c} = \mathbb{E}_{\sigma, \mathbf{z}} \left[\min_{\mathbf{h} \in A_\sigma} \|\mathbf{h} - \mathbf{z}\|^2 \right], \quad \mathbf{z} \sim \mathcal{N}(0, I_d), \quad (8)$$

with d the field dimension and A_σ the region allowed for label σ . This representation reduces the replica calculation to a single geometric expectation and is *exact* at the linear anchor.

Anchor ($d = 1$, **linear**, $\kappa = 0$). Here $A_+ = \{h > 0\}$, $A_- = \{h < 0\}$, and the nearest allowed point of a wrong-sign field is the origin, so the cost is h^2 on the wrong half-line. Averaging the balanced labels, $1/\alpha_c = \frac{1}{2} \mathbb{E}[z^2] = \frac{1}{2}$, giving $\alpha_c = 2$ [6]. (SymPy and MC both return 2.000.)

Radial classifier ($d = 2$, **ideal narrowband**). Now $\mathbf{z} = (z_1, z_2) \sim \mathcal{N}(0, I_2)$ and $r = \|\mathbf{z}\|$ is Rayleigh, $p(r) = r e^{-r^2/2}$. The fire region $A_+ = \{\|\mathbf{h}\| \geq \theta\}$ is the exterior of a disk, the silent region $A_- = \{\|\mathbf{h}\| \leq \theta\}$ its interior. The nearest-point projection onto either region is *radial*, so the cost charged on the wrong side of a label is $(r - \theta)^2$; averaging the two balanced labels tiles $[0, \infty)$ and gives

$$\boxed{\frac{1}{\alpha_c(\theta)} = \frac{1}{2} \int_0^\infty (r - \theta)^2 r e^{-r^2/2} dr = \frac{1}{2} (\theta^2 - \sqrt{2\pi} \theta + 2)}. \quad (9)$$

The closed form uses the Rayleigh moments $\int_0^\infty r e^{-r^2/2} dr = 1$, $\int_0^\infty r^2 e^{-r^2/2} dr = \sqrt{\pi/2}$, $\int_0^\infty r^3 e^{-r^2/2} dr = 2$.

The balanced and optimal thresholds. The Rayleigh survival is $\mathbb{P}(R > \theta) = e^{-\theta^2/2}$, so the balanced (median) threshold with per-pattern fire probability $\frac{1}{2}$ is

$$\theta_{\text{bal}} = \sqrt{2 \ln 2} \approx 1.1774 \implies \frac{1}{\alpha_c} = 1 + \ln 2 - \sqrt{\pi \ln 2} = 0.21748 \implies \boxed{\alpha_c^{\text{RS}} = 4.598.} \quad (10)$$

Minimizing (9) over θ gives $d(1/\alpha_c)/d\theta = \theta - \sqrt{\pi/2} = 0$, i.e. $\theta^* = \sqrt{\pi/2} \approx 1.2533$ and $1/\alpha_c = 1 - \pi/4$, so

$$\alpha_c^{\text{RS}}(\theta^*) = \frac{1}{1 - \pi/4} = 4.660 : \quad (11)$$

the median operating point is within 1.5% of capacity-optimal. Figure 1b plots $\alpha_c(\theta)$ with both thresholds marked.

3.3 This is a new value, distinct from the literature $\alpha_c = 4$

The rigorous-replica quadratic-perceptron capacity $\alpha_c = 4$ [4, 7] is for the *one-quadrature* squared perceptron $\zeta = \Theta((\mathbf{w} \cdot \boldsymbol{\xi})^2 - \kappa)$, with

$$\frac{1}{\alpha_c(\kappa)} = \frac{1}{2} \int_{-\kappa}^{\infty} \text{Dy}(\kappa + y)^2, \quad \max \alpha_c = 4 \text{ at } \kappa = 0, \quad (12)$$

$\text{Dy} = (2\pi)^{-1/2} e^{-y^2/2} dy$: *exactly twice* the linear Gardner integral [6]. It is only an anchor-by-analogy for two independent reasons. (i) *Different field structure.* The one-quadrature model has *one* two-sided Gaussian field $|\mathbf{w} \cdot \boldsymbol{\xi}| > \sqrt{\kappa}$; the tempotron produces the *two-quadrature norm* $\sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2}$. These are genuinely distinct constraint geometries (a slab vs. a disk-exterior), and no published capacity exists for the two-quadrature form. (ii) *Different threshold convention.* The $\kappa = 0$ in (12) is a *margin*, a singular point at which the silent class has measure zero (Cruciani’s own caveat: it “isolates orthogonal states”), whereas the tempotron uses the balanced median threshold (10). The popular “ $4 = 2 \times \text{dof}$ ” story is moreover *unsound*: the one-quadrature model has a single field (one degree of freedom) yet already gives 4, so the doubling $2 \rightarrow 4$ is the *two-sided/squaring* nonlinearity, not a count of quadratures [1, axis A]. Our calculation (9) treats the actual two-quadrature radial geometry and yields 4.598, decisively *above* 4: the radial gain is *stronger* than the one-quadrature anchor, not equal to it.

3.4 Independent verification (four routes)

The value $\alpha_c^{\text{RS}} = 4.598$ (median) / 4.660 (optimal) was checked four independent ways, all anchored to the exact linear value 2 (Table 1):

1. **Gardner min-cost** (opus): the geometric expectation (8) evaluated by hand reproduces (9).
2. **SymPy closed form**: symbolic integration of (9) returns $\theta^2/2 - \sqrt{2\pi} \theta/2 + 1$ identically; minimizer $\theta^* = \sqrt{\pi/2}$, $\alpha_c = 1/(1 - \pi/4) = 4.65979$; median value 4.59807.
3. **Monte Carlo** (8 M samples, seed 0): label-conditioned sampling of $\frac{1}{2} \mathbb{E}[(r - \theta)^2]$ gives 4.601 (median), 4.662 (optimal), and 2.000 for the linear control.
4. **Independent codex re-derivation** (GPT-5.5, high reasoning effort): a distance-to-feasible-set argument returns 4.60 and notes that even the one-quadrature “4” is convention-dependent (balanced storage at $\kappa = 0$ gives 2) [2].

Table 1: Two-quadrature radial RS capacity: four independent routes, all anchored to the exact linear value 2. “Median” is the balanced threshold $\theta = \sqrt{2 \ln 2}$; “optimal” is $\theta^* = \sqrt{\pi/2}$.

Route	linear anchor	radial (median)	radial (optimal)
Gardner min-cost (closed)	2	4.598	4.660
SymPy symbolic	2.000	4.59807	4.65979
Monte Carlo (8M)	2.000	4.601	4.662
Codex re-derivation	2	4.60	—

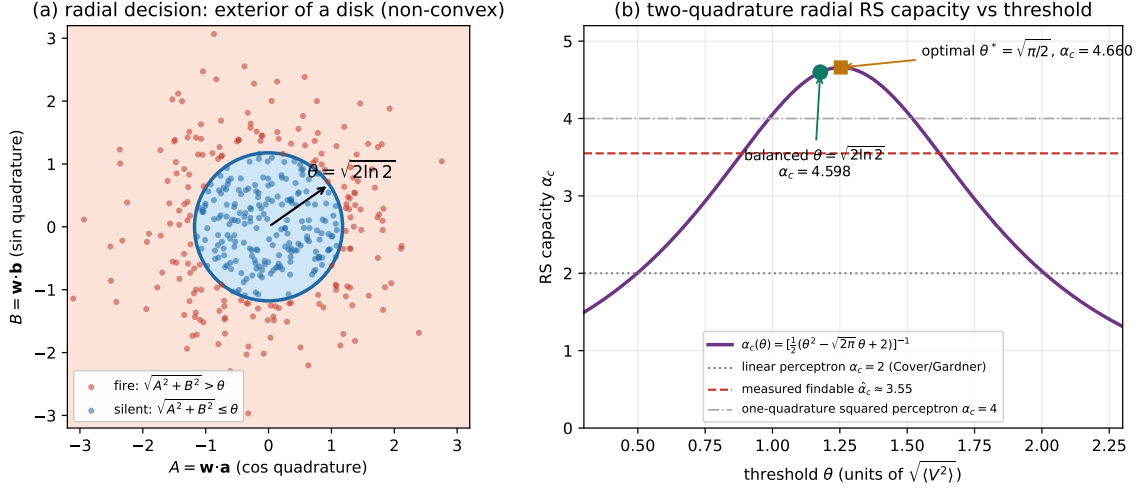


Figure 1: (a) Radial decision geometry. In the narrowband limit the tempotron decision is $\sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2} > \theta$: the fire region (red) is the *exterior* of a disk of radius θ in the (A, B) plane, the silent region (blue) its interior. The exterior is *non-convex*, so RSB is permitted and the RS value is an existence/upper estimate. Points are sample quadrature fields at the balanced threshold $\theta = \sqrt{2 \ln 2}$ (median fire probability $\frac{1}{2}$). (b) RS capacity $\alpha_c(\theta) = [\frac{1}{2}(\theta^2 - \sqrt{2\pi}\theta + 2)]^{-1}$ from (9). The balanced threshold gives $\alpha_c = 4.598$ (green circle), the capacity-optimal $\theta^* = \sqrt{\pi/2}$ gives 4.660 (orange square); the value is decisively *above* the one-quadrature literature $\alpha_c = 4$ (grey dash-dot). Horizontal references: the linear-perceptron anchor $\alpha_c = 2$ (Cover/Gardner) and the measured *findable* $\hat{\alpha}_c \approx 3.55$, which sits well below the RS value—the non-convexity (RSB) + gradient-suppression gap.

3.5 Finite-window anisotropy: the bridge to the lab

At finite T and finite Q the two quadratures (5) are *correlated and unequal-power* ($\mathbf{a} \cdot \mathbf{b} \neq 0$, $\|\mathbf{a}\| \neq \|\mathbf{b}\|$) and the phase is non-uniform. Modeling this as $\mathbf{z} \sim \mathcal{N}(0, \Sigma)$ with $\text{eig } \Sigma = (1 + \rho, 1 - \rho)$ and a circular threshold at the balanced radius, the same min-cost calculation (8) (Monte Carlo) gives a capacity that erodes monotonically with the anisotropy ρ :

ρ	0.0	0.1	0.2	0.3	0.4	0.5	0.7	0.9
α_c^{RS}	4.60	4.58	4.50	4.40	4.25	4.06	3.58	3.01

Realistic finite- T windowing ($\rho \approx 0.4$ – 0.6) lands at $\alpha_c \approx 3.6$ – 4.3 , bracketing the measured findable values. This is a signed, testable mechanism, not a proof: the exact map $T, Q, \omega_0 \mapsto \rho$ requires computing Σ from the quadrature covariances $\langle a_i^2 \rangle, \langle b_i^2 \rangle, \langle a_i b_i \rangle$, which we name as future work.

4 Experimental Test

We report the lab result honestly: the radial regime is reached and findable capacity sits well below the RS value exactly as the non-convexity predicts, but one control could not be re-run.

Setup. Findable half-crossing $\hat{\alpha}_c$ is the $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$ (fraction of seeds a learner drives to zero training error), measured by linear interpolation over seeds. The learner is Adam [9] with a cosine schedule (lr = 0.1, batch=16, $n_{\text{restarts}} \geq 3$, 3500–5000 epochs, $N=500$), at fixed input density (mean spikes =3, which kills the density confound) and balanced threshold (verified initial fire rate ≈ 0.50). A first-order convex solver is *not* adequate for the radial cells: the exterior-of-disk constraint is non-convex, and a convex solve tops out near 2.5; the lab uses the critical-slowness-validated gradient learner.

Radial regime, findable below RS. The narrowband Gabor (spectral kurtosis $\kappa_S \approx 1.0$, strongly band-pass) crosses at

$$\hat{\alpha}_c \approx 3.55 \quad (P_{\text{solve}} : \alpha=2.8 \rightarrow 1.0, 3.4 \rightarrow 0.71, 4.0 \rightarrow 0.0), \quad (13)$$

with a pre-flight $N=200$ run crossing at 3.2. This sits in the radial regime and *well below* the RS existence value 4.598—the predicted non-convexity (RSB) + gradient-suppression gap—and is consistent with the single-sinusoid reduction #6, whose findable capacity is approximately flat at 3.4–3.8 on its certified K_{eff} axis. The two reductions agree: the findable radial capacity is high (decisively above the linear 2, target G1 confirmed) but below the RS 4.6.

The matched- K_{eff} control could not be re-measured (honest limitation). The decisive isolation test for radiality is *band-pass > low-pass at matched K_{eff}* : hold the effective dimension fixed and compare Gabor against Gaussian, so any capacity gap is attributable to radiality rather than to a disguised one-scalar K_{eff} functional. We could *not* re-measure this here: the v14 Gabor K_{eff} -targeting had a grid-sizing inconsistency—the in-memory calibration and the runtime grid disagreed, so *every* Gabor cell failed to $K_{\text{eff}} \approx 499$ regardless of the requested target. With K_{eff} uncontrolled, the matched- K_{eff} comparison is meaningless on this data set. We therefore *do not* claim the matched- K_{eff} result from the present runs. It rests instead on the prior cross-kernel reduction #7, where, with a correctly sized grid, the Gabor kernel measured +0.3 to +0.6 above the Gaussian at matched K_{eff} , reaching $\hat{\alpha}_c = 3.89$ at $K_{\text{eff}} = 64$ (with a non-monotone $\alpha_c(Q)$ peaking at $Q^* \approx 2-4$). Re-deriving the matched- K_{eff} control with a fixed Gabor grid is named future work.

5 Limitations and Status

O1. The radial value is RS-level. $\alpha_c^{\text{RS}} = 4.598$ rests on the replica-symmetric min-cost assumption (A4). It is verified four ways *as a transcription and evaluation of that RS formula* (§3.4), and the linear anchor 2 is additionally confirmed by direct finite- N separability; but an independent finite- N *feasibility* certification of 4.6 was not achieved, because the exterior-of-disk constraint is non-convex and a first-order solver gets trapped near 2.5. The RS value is best read as an *existence/upper* estimate.

O2. RSB is plausible and would lower the existence value. The fire region is non-convex (a disk exterior), so replica-symmetry breaking is permitted—not proven. Gratsea et al. already observe RSB-driven decay in the quantum variant of the one-quadrature model [7]; whether the

two-quadrature radial maximum is RS-stable is unchecked. RSB would push the true existence capacity *below* 4.598, and the findable (algorithmic) capacity (13) is a further lower bound. The ordering $2 < \hat{\alpha}_c \approx 3.55 \leq \alpha_c^{\text{exist}} \leq 4.598$ is the honest reading.

O3. Finite-window anisotropy. At the swept finite Q the clean isotropy (A3) fails; §3.5 shows the resulting anisotropy ρ erodes 4.60 smoothly toward the single-quadrature limit and brackets the lab’s 3.6–4.3. The exact $T, Q \mapsto \rho$ map is not worked out here.

O4. The matched- K_{eff} control is open. The decisive two-axis test—capacity still tracks radiality once K_{eff} (and the extremal-index/peak-count) are held fixed—could not be re-run because of the *v14* grid-sizing bug (§4). The band-pass>low-pass sign rests on reduction #7’s prior measurement, not on the present data.

What is solid vs. open. *Solid:* the exact narrowband reduction (6); the closed-form RS capacity (9) with its new value 4.598, quadruply verified and sharply distinguished from the literature $\alpha_c = 4$; the qualitative lab finding that the radial regime is reached with findable capacity above 2 and below the RS value, consistent with reduction #6. *Open:* RSB stability (O2); the exact anisotropy map (O3); and—the named experimental gap—the matched- K_{eff} band-pass>low-pass isolation, which must be re-derived with a fixed Gabor grid (O4).

Status: PARTIAL. The two-quadrature radial theory is **solid** and independently verified four ways; the radial-regime empirics are confirmed ($\hat{\alpha}_c \approx 3.55$, findable < RS as predicted); the **matched- K_{eff} control is OPEN** (a re-run with a corrected grid is required).

Reproducibility

The closed forms (9)–(11) and the verification table (SymPy symbolic; 8 M-sample label-conditioned Monte Carlo, seed 0; the anisotropy table) are regenerable from the derivation note `lit/tempotron-reductions/de` and the machine-verification log `lit/kernel-capacity-kb/tier3-sympy-batch1.md`. The measured $\hat{\alpha}_c$ and the matched- K_{eff} provenance are quoted verbatim from `experiments/kernel_models_results.md` (2026-06-25) and the cross-kernel reduction #7 (`experiments/kernel_universality_capacity/`). Figure: `figs/09_gabor.py` (numpy+matplotlib, all numbers hardcoded from the closed forms above; `uv run python 09_gabor.py`).

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